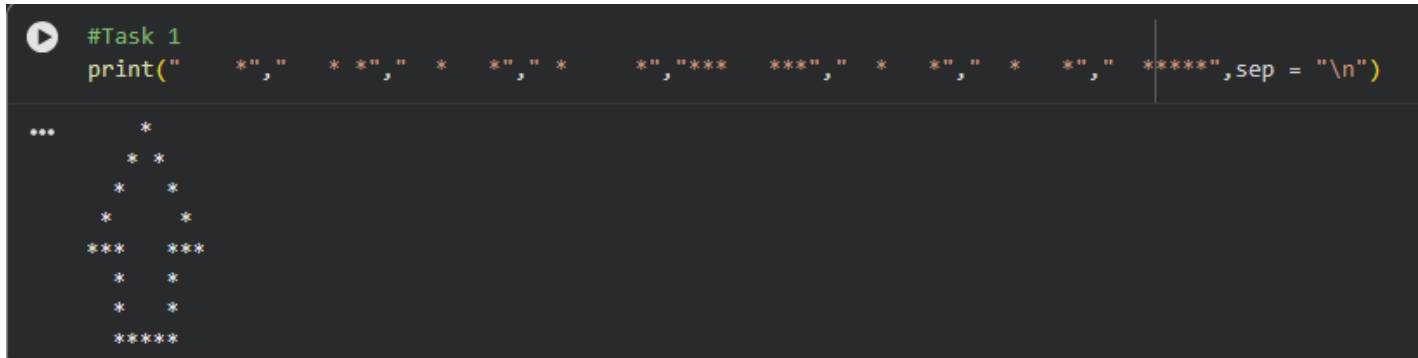


Hands-on Activity 1.1	
Python Data Types, Variables, Operators, and Basic I/O Operations	
Course Code: CPE009A	Program: BSCPE
Course Title: Object Oriented Programming 1	Date Performed:12/17/25
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Name(s):Condino,Niel Vincent B.	Instructor: Engr. Jimlord Quejado

6. Output

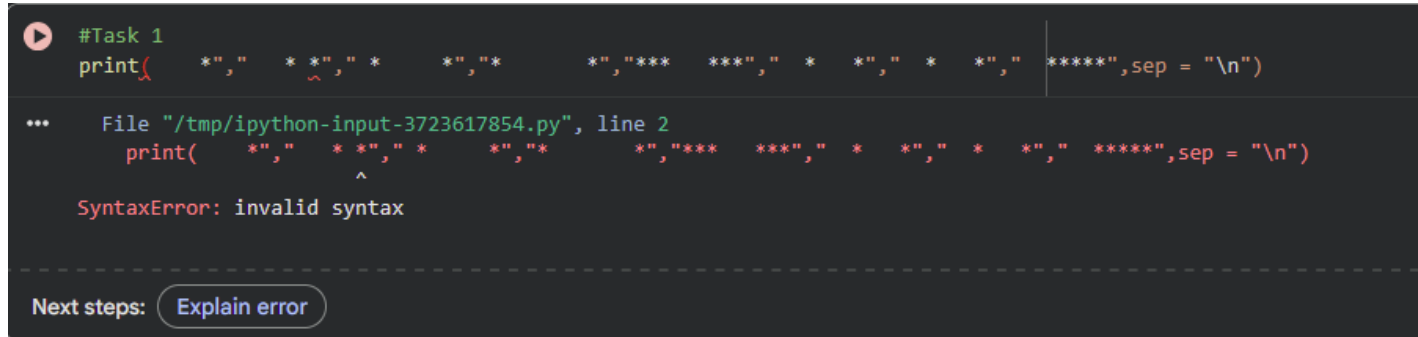


```
#Task 1
print(" *", " * *", " * * *", " * * * *", " * * * * *", sep = "\n")

...
 *
 * *
 * * *
 * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
```

Figure 1. Screenshot of “Task 1”’s code and output

The figure above shows the accomplishment on task 1. In this code, the code that will span different lines is compressed to just one line using the separator “\n”. The output shows the same arrow as it will using the original code.



```
#Task 1
print( *", " * *", " * * *", " * * * *", " * * * * *", sep = "\n")

...
File "/tmp/ipython-input-3723617854.py", line 2
    print( *", " * *", " * * *", " * * * *", " * * * * *", sep = "\n")
           ^
SyntaxError: invalid syntax

Next steps: Explain error
```

Figure 1.1. Screenshot of “Task 1”’s invalid syntax

The figure above shows the error message due to removing one of the quotation marks. This causes a syntax error in python that stops it from compiling and running the program.


```

#Task 4
# input a float value for variable a here # input a float value for variable b here
a = float(input("Input a number for variable a: "))
b = float(input("Input a number for variable b: "))

print("\nThe sum of a and b is:",a+b)
print("\nThe difference of a and b is:",a-b)
print("\nThe product of a and b is:",a*b)
print("\nThe quotient of a and b is:",a/b)
print("\nThat's all, folks!")

Input a number for variable a: 52
Input a number for variable b: 66

The sum of a and b is: 118.0

The difference of a and b is: -14.0

The product of a and b is: 3432.0

The quotient of a and b is: 0.7878787878787878

That's all, folks!

```

Figure 4. Screenshot of "Task 4"s code and output

Figure 4 presents the code of a program that computes all the arithmetic answers for 2 inputted numbers. It first asks for a value for a and b which will be the 2 numbers. These values are both converted into a float. After this, a series of print functions is used. The first displays the sum. The next one displays the difference. Next, the product is displayed. After this, the quotient of a and b is displayed. All of the calculated answers are computed using the arithmetic operations in python..Lastly it prints a farewell message.

7. Supplementary Activity

```

#Supplementary 1
print("=====Triangle Area Calculator=====|")

base = float(input("Input base: "))
height = float(input("Input height: "))

area = (base*height) / 2

print("Area of the triangle is", area,"units squared")

... =====Triangle Area Calculator=====
Input base: 5
Input height: 10
Area of the triangle is 25.0 units squared

```

Figure 5. Screenshot of "Supplementary 1"s code and output

Figure 5 shows a code of a program that calculates the area of a triangle. The program first asks for the value for base and height of the triangle. These values are then converted into a float. The area is calculated using the formula of the product of base and height divided by 2.

```

#Supplementary 2
print("===== Celsius to Farenheit Converter =====")

cTemp = float(input("Input the temp in Celsius: "))

fTemp = (cTemp * (9/5)) + 32

print(cTemp, "in Degrees Celsius is",fTemp,"in Degrees Farenheit")

... ===== Celsius to Farenheit Converter =====
Input the temp in Celsius: 10
10.0 in Degrees Celsius is 50.0 in Degrees Farenheit

```

Figure 6. Screenshot of “Supplementary 2”’s code and output

Figure 6 shows a code in python that converts celsius to fahrenheit. The program first asks for an input of temperature in celsius. This input is then converted into a float. The fahrenheit equivalent of the temperature is then calculated using the formula : $(C * 9/5) + 32$. Lastly, the program displays the input temperature and its converted temperature

```

#Supplementary 3
print("===== Distance Calculator =====")

x1 = float(input("Input x1: "))
x2 = float(input("Input x2: "))
y1 = float(input("Input y1: "))
y2 = float(input("Input y2: "))

d = (((x2 - x1)**2) + ((y2-y1)**2))**0.5

print("distance between coordinates is", d)

... ===== Distance Calculator =====
Input x1: 3
Input x2: 6
Input y1: 4
Input y2: 8
distance between coordinates is 5.0

```

Figure 7:Screenshot of “Supplementary 3”’s code and output

Figure 7 shows a code that calculates a distance between 2 points. The program first asks for the values of x1,x2,y1, and y2. These inputs are converted into a float. The value of the distance is calculated then using the square root of the quantity of the difference of x2 and x1 squared added to the difference of y2 and y1 squared. This value is stored inside the variable d. After this the program prints the value of d.

8. Conclusion

In this activity, I am able to familiarize myself more with python's syntax. Python and C++ has a lot of commonalities as they are both programming languages. They both share the concept of variable, arithmetic operation, literals or data type, etc. . As I observe, python is a lot more forgiving than C++ when it comes to its syntax. The different tasks in this lab are just a warm-up and experimenting on python. As stated earlier, because both of the languages share the same concepts, the activities are quite easy enough to accomplish as these are still the basics. The simplified syntax also of python made transferring the concept from another language easier. The supplementary activities are more focused on the arithmetic operation of python. The basic operations on C++ also appear here but there are new additions like a built-in exponential operation and division being separated to floor division for increased precision and more beginner friendly experience. These tasks also helped me understand more about translating math formulas to the python language. Overall, as this is the first activity, these tasks are basic and easy to do as it is just the introductory lesson for python. I expect the next lessons to teach more about different concepts in python such as functions, especially the list and dictionaries as these concepts are not built-in in C++ as far as I know.

9. Assessment Rubric